Multi-scale search is still search

Effective trials = N x k, not N - the frame axis spends significance too
auditkit v0.2.0 - 2026-11-03 - MIT
Scope: a count of what you looked at, not proof that the edge is fake

METHOD

The deflated-Sharpe bar is a best-of-N bar: it asks whether this Sharpe beats the best of N tries. If you tuned the same grid on k time frames, every frame is another try. The bar grows like sqrt(2 ln N_eff / T), so the honest count is N_eff = N x k.

WORKED EXAMPLE - T = 400 periods, per-period Sharpe 0.11

Trial bookkeeping	N	Bar	Adj. p	Verdict
2 configs, one frame	2	0.059	0.028	PASS
2 configs x 10 frames	20	0.122	0.28	FAIL

READING

The strategy did not change. Only the count of things you looked at changed. If the frame was pre-registered - chosen before you saw the data - then k = 1 and nothing is added. The rule counts searches, not mentions of time.

REPRODUCE

```
python3 -m auditkit.cli trades.csv --trials 100 --timeframes 5 JSON adds timeframes,
trials_effective, multiscale_inflation.
```

SCOPE AND LIMITS

Frames on the same price series are correlated, so N x k overstates the number of independent trials. We report the auditable count, not the flattering one.
Inherits the assumptions of the underlying check: i.i.d. periods, normal approximation, Sharpe as the summary statistic.
Does not detect the case where frames were selected after seeing results - only you can report that.